Semester One Examination, 2013

Question/Answer Booklet 1

1

CHEMISTRY 3AB

Student name: ______________________________

Teacher: ______________________________

Time allowed for this paper

Reading time before commencing work:	ten (10) minutes
Working time for paper:	three (3) hours

Materials required/recommended for this paper

To be provided by the supervisor

Three (3) Question/Answer booklets
Multiple-choice Answer Sheet attached to Booklet 3
Chemistry Data sheet

To be provided by the candidate

Standard items:	pens, pencils, eraser, correction fluid/tape, ruler, highlighters
Special items:	non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-choice	25	25	50	**25**	25
Section Two: Short answer	11	11	60	**70**	35
Section Three: Extended answer	6	6	70	**80**	40
			Total	**175**	100

Instructions to candidates

1. Write your name and your teacher's name on **all three (3**) booklets.

2. The rules for the conduct of Western Australian external examinations are detailed in the *Student Information Handbook 2013*. Sitting this examination implies that you agree to abide by these rules.

2. Answer the questions according to the following instructions.

   Section One: Answer all questions on the Multiple-choice answer sheet attached to the front of Booklet 3. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

   Sections Two and Three: Write your answers in the Question/Answer Booklets provided.

3. When calculating numerical answers, show your working or reasoning clearly unless instructed otherwise.

4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

5. Spare pages are included at the end of all Booklets. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages in planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
     Fill in the number of question(s) that you are continuing to answer at the top of the page.

Section One: Multiple-choice **25% (50 marks)**

This section has **25** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 50 minutes.

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1. Which of the following correctly lists the elements in the order of **increasing** radius?

(A) Br < Se < As < Ge

(B) Ge < As < Se < Br

(C) As < Se < Br < Ge

(D) Se > Br > Ge < As

2. The first four successive ionisation energies for element **X** are

0.637 MJmol^{-1} ***1.24 MJmol^{-1}*** ***2.41 MJmol^{-1}*** ***7.12 MJmol^{-1}***

Which of the following formulae is most likely for the oxide of element **X**?

(A) XO

(B) X_2O

(C) XO_3

(D) X_2O_3

3. Which one of the following statements about Group 17 elements is **true**?

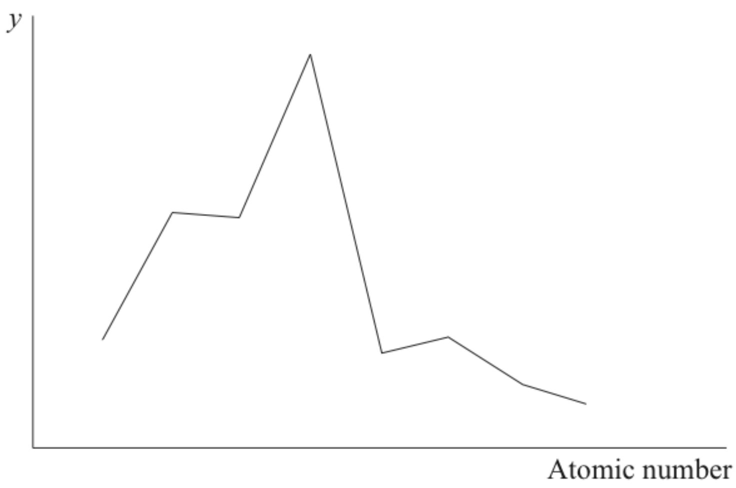
(A) The radius of chloride ion is larger than that of the iodide ion.

(B) The boiling point of liquid bromine (Br_2) is higher than that of liquid fluorine (F_2).

(C) An iodine atom is more electronegative than a chlorine atom.

(D) The first ionisation energy of bromine is higher than that of fluorine.

4. The x-axis of the graph below represents the atomic number of the elements in period 3.



Which variable could represent the **y**-axis?

(A) Melting point

(B) Electronegativity

(C) Ionisation energy

(D) Atomic radius

5. Which statements about the periodic table are **correct**?

I. The elements Mg, Ca and Sr have similar chemical properties.
II. Elements in the same period have the same number of electron shells.
III. The oxides of Na, Mg and P are basic.

(A) I and II only

(B) I and III only

(C) II and III only

(D) I, II and III

6. Which of the following describes the nature of the covalent bonds and shape of the OF_2 molecule?

(A) non-polar and linear

(B) polar and bent

(C) non-polar and bent

(D) polar and linear

7. Which of the following statements is **not true** of van der Waals' forces?

(A) They are all intermolecular interactions.

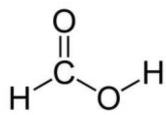
(B) They include dispersion forces which are of significance between non-polar molecules in both the liquid and solid phase.

(C) They are not significant in covalent network substances since covalent bonding is much stronger than van der Waals forces.

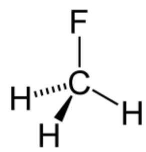
(D) They account for the stability of nitrogen gas molecules (N_2) which can not be broken down into separate nitrogen atoms even at high temperature.

8. In which of the following liquids would you expect hydrogen bonding to be present between their molecules

Hydrogen fluoride



Methanoic acid



Fluoromethane

(A) hydrogen fluoride and methanoic acid only

(B) hydrogen fluoride and fluoromethane only

(C) all of them

(D) hydrogen fluoride only

9. The equation for an equilibrium system involving hydrogen iodide is as follows

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)} \qquad \Delta H = -51.8 \text{ kJ mol}^{-1}$$

Which of the following changes would cause the rate of formation of products to increase?

I. Increasing the concentration of H_2.

II. Increasing the concentration of HI

III. Increasing the volume of the system without changing the amount of substance present.

IV. Increasing the temperature.

(A) II and IV only

(B) II and III and IV only

(C) II and III only

(D) I, II and IV only

10. When chloride ions are added to a solution containing $Co(H_2O)_6^{2+}{}_{(aq)}$, the following equilibrium is established.

$$Co(H_2O)_6^{2+}{}_{(aq)} \;\; 4Cl^-{}_{(aq)} \rightleftharpoons CoCl_4^{2-}{}_{(aq)} + 6H_2O_{(l)}$$

Pink Blue

Which of the following statements about the colour of the solution is **true**?

(A) Diluting the solution with water will make it turn blue.

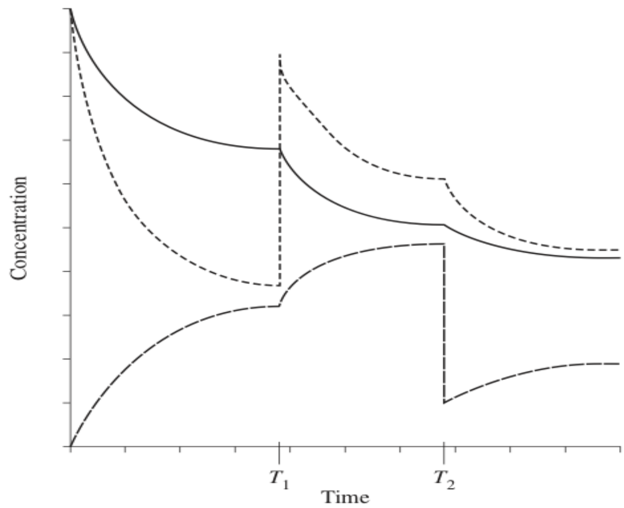
(B) If the reaction is exothermic, heating the solution will make it turn blue.

(C) If the reaction is endothermic, cooling the solution will make it turn pink.

(D) Adding a large amount of solid potassium chloride to the solution will make it turn pink.

11. The graph shows the concentrations over time for the system:

$$CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$$



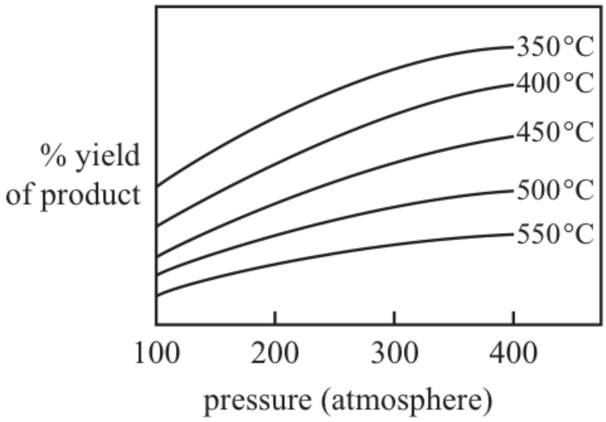
What happens at time T_1 and T_2 ?

	T_1	T_2
(A)	H_2 added	CH_3OH removed
(B)	CO added	CH_3OH removed
(C)	H_2 added	CO removed
(D)	CO added	CO and H_2 removed

Questions 12 and 13 refer to the following information.

The graph below refers to the gaseous equilibrium reaction: $\mathbf{CO_{2(g)} + H_{2(g)} \rightleftharpoons CO_{(g)} + H_2O_{(g)}}$

The effect of increasing pressure and temperature on the yield is shown in the graph below.



12. Which one of the following statements about the above gaseous reaction is **true**?

(A) The reaction is endothermic because the yield increases as the temperature increases.

(B) The reaction is exothermic because the yield increases as the temperature increases.

(C) The reaction is exothermic because the yield decreases as the temperature increases.

(D) The reaction is endothermic because the yield decreases as the temperature increases.

13. Which of the following **best** describes the effect of a catalyst on the reaction above?

	Effect on the rate of the forward reaction	*Effect on the rate of the reverse reaction*	*Effect on equilibrium*
(A)	increase	increase	favours forward reaction
(B)	increase	increase	favours reverse reaction
(C)	increase	increase	no change
(D)	increase	decrease	no change

14. Consider a solution of lead (II) chloride at 25°C in equilibrium with some solid lead (II) chloride at the bottom of the solution.

$$Pb^{2+}_{(aq)} + 2Cl^-_{(aq)} \rightleftharpoons PbCl_{2(s)}$$

Which of the following would be the **best** way to increase the concentration of the lead and chloride ions in the solution?

(A) Allowing some of the water to evaporate and then leaving the system to reach equilibrium again.

(B) Heating the system to 30°C.

(C) Adding some more solid lead(II) chloride.

(D) Sealing the container and vigorously shaking the solution.

15. Under the conditions used industrially, ethene and steam react as follows

$$C_2H_{4(g)} + H_2O_{(g)} \rightleftharpoons C_2H_5OH_{(g)} \qquad \Delta H = -46 \text{ kJ mol}^{-1}$$

Which set of conditions would give the **best** yield of ethanol at equilibrium?

(A) High temperature, low pressure

(B) High temperature, high pressure

(C) Low temperature, high pressure

(D) Low temperature, low pressure

16. The following sets of solutions are mixed. In which of these will there be no visible reaction?

(A) Set 1 consisting of barium hydroxide, dilute sulfuric acid, potassium nitrate

(B) Set 2 consisting of potassium phosphate, copper (II) sulfate and sodium chloride

(C) Set 3 consisting of sodium chloride, copper (II) sulfate and silver nitrate

(D) Set 4 consisting of sodium iodide, barium nitrate and potassium hydroxide

17. When charcoal (C) reacts in the presence of oxygen, carbon monoxide and carbon dioxide are produced according to the following chemical reactions:

$$C_{(s)} + \tfrac{1}{2} O_{2(g)} \rightarrow CO_{(g)}$$

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$$

What would be the total mass of gas produced when 400 g of charcoal is reacted, assuming equal amounts are consumed in each reaction?

(A) 0.93 kg

(B) 1.20 kg

(C) 1.50 kg

(D) 2.50 kg

18. All of the carbon dioxide in a soft drink with an initial mass of 381.04 g was carefully extracted and collected as a gas. The final mass of the drink was 380.41 g.

What volume would the carbon dioxide occupy at 100 kPa and 25°C?

(A) 0.335 L

(B) 0.355 L

(C) 0.565 L

(D) 0.635 L

19. All the lead ions present in a 50.0 mL solution were precipitated by reaction with excess chloride ions. The mass of the dried precipitate was 0.595 g.

What was the concentration of lead in the original solution?

(A) $8.87\ g\ L^{-1}$

(B) $10.2\ g\ L^{-1}$

(C) $11.9\ g\ L^{-1}$

(D) $16.0\ g\ L^{-1}$

20. Chromium carbonate is insoluble and precipitates according to the chemical equation below

$$2\ Cr^{3+}_{(aq)} + 3\ CO_3^{2-}{}_{(aq)} \rightarrow Cr_2(CO_3)_{3(s)}$$

A solution containing 0.600 mol of chromium nitrate is added to a solution containing 0.600 mol of sodium carbonate. How much chromium carbonate is precipitated?

(A) 0.200 mol

(B) 0.300 mol

(C) 0.400 mol

(D) 0.600 mol

21. Pure water at 100 °C has a pH of 6.14. This is because

(A) pH measurements at this temperature are unreliable.

(B) pH measurements are affected by the bubbles of hydrogen gas that form in boiling water.

(C) the self-ionisation of water is endothermic.

(D) the concentration of H_3O^+ ions is not equal to the concentration of OH^- ions at this temperature.

22. Which of the following are examples of Lowry-Bronsted acid-base reactions?

I. $Mg + 2\ H^+ \rightarrow Mg^{2+} + H_2$

II. $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$

III. $NH_2^- + H_2O \rightarrow NH_3 + OH^-$

IV. $3\ Cu + 8\ H^+ + 2\ NO_3^- \rightarrow 3\ Cu^{2+} + 2\ NO + 4\ H_2O$

(A) I, II and III only

(B) I and II only

(C) II, III only

(D) II, III and IV only

23. A 0.100 $molL^{-1}$ ethanoic solution is 1.34% ionised at a fixed temperature. The pH of the solution is

(A) 3.41

(B) 2.87

(C) 1.34

(D) 0.870

24. Which of the following equations **best** represents the chemical reaction between a 0.1 $molL^{-1}$ sodium hydroxide solution and dilute ethanoic acid solution?

I. $CH_3COOH_{(aq)} + OH^-_{(aq)} \rightarrow CH_3COO^-_{(aq)} + H_2O_{(l)}$

II. $CH_3COOH_{(aq)} + K^+_{(aq)} + OH^-_{(aq)} \rightarrow CH_3COO^-_{(aq)} + H_2O_{(l)} + K^+_{(aq)}$

III. $CH_3COOH_{(aq)} + KOH_{(aq)} \rightarrow KCH_3COO_{(aq)} + H_2O_{(l)}$

IV. $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$

(A) Both I and IV are equally correct

(B) Both I and III are equally correct.

(C) I only

(D) IV only

25. A acid-base buffer is a solution that resists change in pH. Which of the following mixtures would make a good buffer solution??

(A) 100.0 mL of 0.10 $molL^{-1}$ $NaHSO_4$ solution and 100.0 mL of 0.10 $molL^{-1}$ Na_2SO_4 solution.

(B) 100.0 mL of 0.10 $molL^{-1}$ H_2SO_4 solution and 100.0 mL of 0.10 $molL^{-1}$ $NaHSO_4$ solution.

(C) 100.0 mL of 0.10 $molL^{-1}$ NaH_2PO_4 solution and 100.0 mL of 0.10 $molL^{-1}$ Na_3PO_4 solution.

(D) 100.0 mL of 0.10 $molL^{-1}$ NH_3 solution and 100.0 mL of 0.10 $molL^{-1}$ CH_3COOH solution.

END OF MULTIPLE CHOICE SECTION

Section Two: Short answer **35% (70 Marks)**

This section has **11** questions. Answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three (3) significant figures and include appropriate units where applicable.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

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- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 60 minutes

Question 26 **(11 marks)**

(a) Write the electron configuration for each of the species below. (2 marks)

Species	***Electron configuration***
Sulfide ion	
Aluminium atom	

(b) Complete the table below

All valence electron pairs should be presented either as : or — (9 marks)

Species	***Lewis electron dot diagram***	***Sketch or name of shape***	***Polarity of molecule (polar or non-polar)***
NBr_3			
OBr_2			
SO_3			

Question 27 **(5 marks)**

(a) Place each of the following substances in the appropriate column of the table below on the basis of the experimental observation stated in the table below. (2 marks)

{ $Br_{2(l)}$, $Cu_{(s)}$, $NH_4NO_{2(s)}$, $Fe_{(s)}$ }

Conducts electricity well but does not dissolve in dilute hydrochloric acid	*Boils at a temperature below 100°C*	*Forms a pale green solution when added to a dilute solution of ethanoic acid*	*Can only conduct electricity in molten/aqueous solution*

(b) State the most significant type of intermolecular forces for each of the following liquids. (3 marks)

Liquids	***Most significant intermolecular force***
Sulfur (S_8)	
CH_3F	
H_2SO_4	

Question 28 **(3 marks)**

Write ONE relevant ionic chemical equation to explain each of the observations shown in the table below.

Observation	***Explanatory equation***
The pH of a solution of $NaHCO_3$ is greater than 7.	
The addition of solid $NaNH_2$ to water produces a pungent odour and turns red litmus paper to blue.	
A vinegar smell is produced when a colourless liquid is added to a dilute solution of potassium ethanoate.	

Question 29 **(5 marks)**

Explain the following using relevant concepts and theories.

(a) The first ionization energy of a fluorine atom is higher than that of a boron atom. (3 marks)

(b) The radius of a fluoride ion (F^-) is larger than the radius of sodium ion (Na^+) even though both have the same number of electrons. (2 marks)

END OF BOOKLET 1

Semester One Examination, 2013

Question/Answer Booklet 2

CHEMISTRY 3AB

2

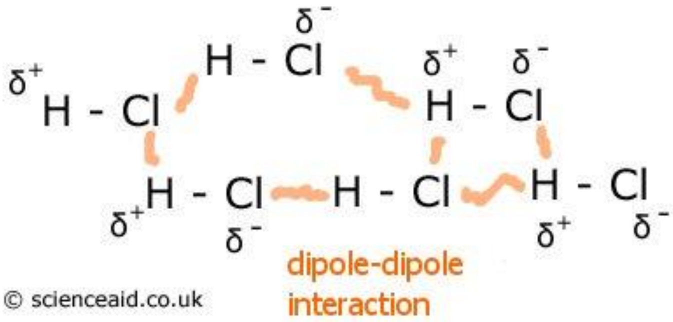
Student name: ______________________________

Teacher: ______________________________

Section Two: Short answer (cont.)

Question 30 **(6 marks)**

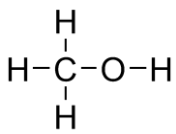
The following diagram shows dipole-dipole interaction in liquid hydrogen chloride.



Liquid methanol (CH_3OH) and solid sodium hydroxide (NaOH) are both soluble in water.

(a) Use the diagram above as a guide to draw relevant diagrams showing the most dominant form of interaction between EACH of the two substances above and liquid water molecules in solution. (Note: an electron dot diagram for methanol has been provided for your reference) (2 marks)

Electron dot diagram for methanol:



methanol in water	*sodium hydroxide in water*
Name of interaction:	*Name of interaction:*

(b) Name the interaction that you have drawn in part (a) in the space provided above. (2 marks)

(c) At atmospheric pressure, methanol has a boiling point of about 65°C while sodium hydroxide has a boiling point of approximately 1390°C. Briefly explain this difference using relevant concepts and theories.

(2 marks)

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Question 31 **(6 marks)**

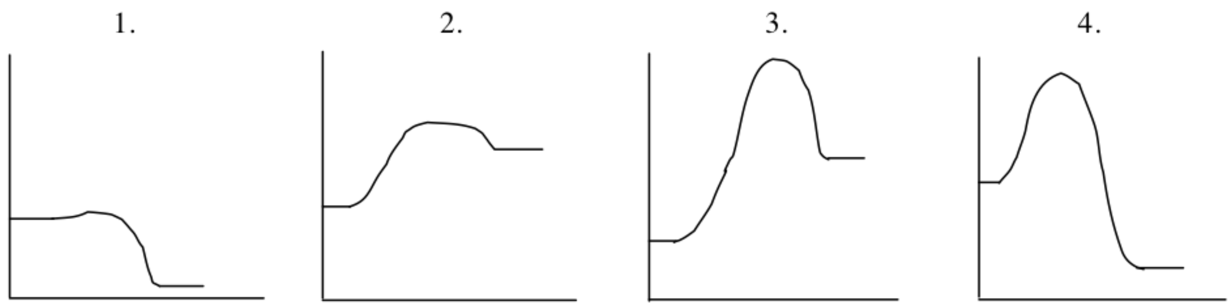
Propane is a hydrocarbon (C_3H_8) and is sometimes referred to as liquefied petroleum gas, LP-gas, or LPG. It is used as an alternative fuel to petroleum and diesel.

People use propane in and around their homes for water heaters, outdoor grills, fireplaces, and appliances. Industrial uses include propane-driven forklifts and fleet vehicles.

The chemical equation for the combustion of propane in air is

$$C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(l)}$$

(a) Consider the four graphs below showing changes in energy (kJ mol^{-1}) against reaction coordinate



One of these graphs represents the combustion of propane. Identify this graph, giving reasons for your choice. Use lines or markings to clearly show activation energy and heat of reaction on your chosen graph. (4 marks)

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(b) Identify the graph that represents a process **most likely** to proceed spontaneously at room temperature. Briefly explain your choice. (2 marks)

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Question 32 **(6 marks)**

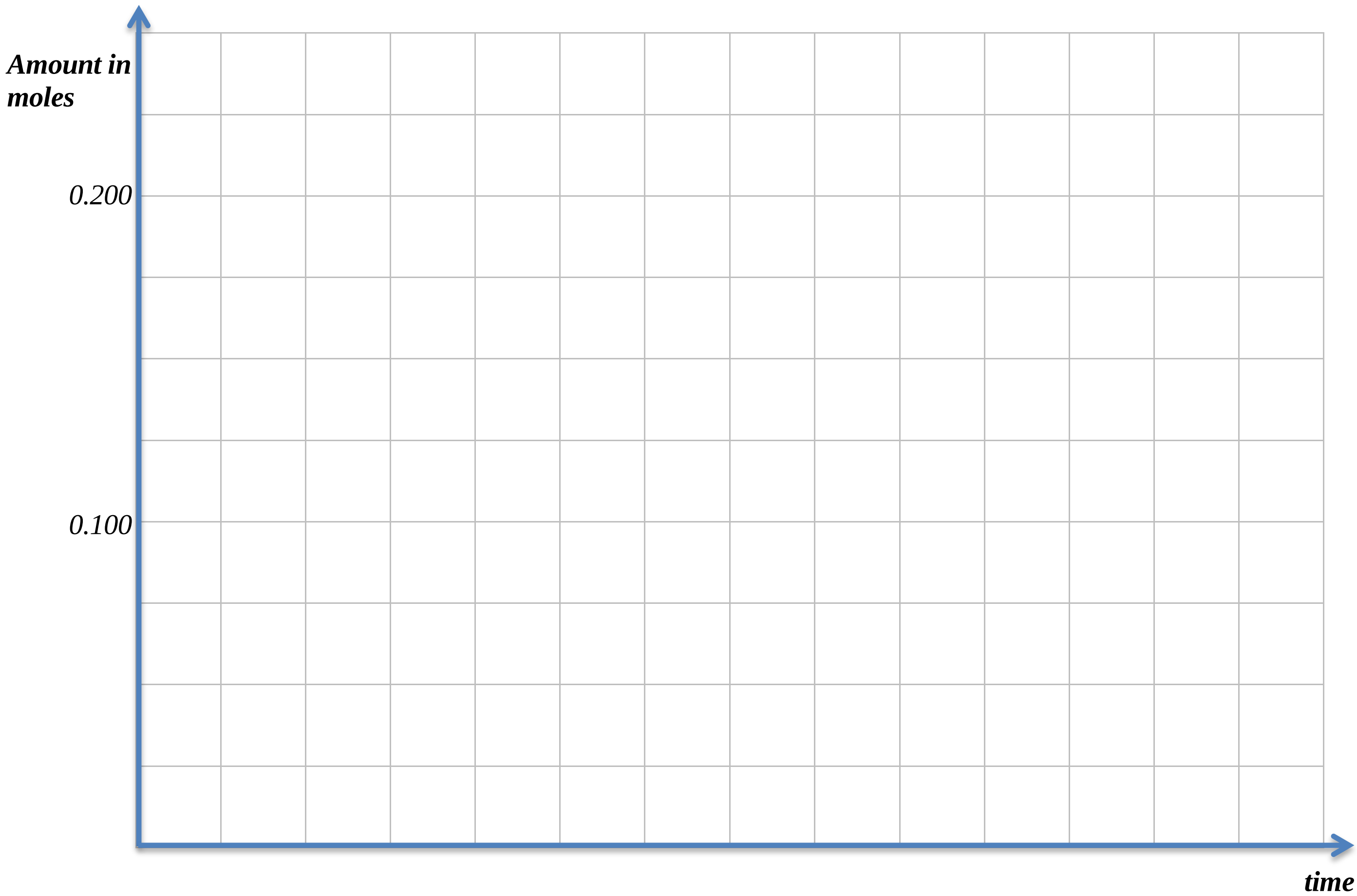
Consider the following reaction:

$$Na_2CO_{3(s)} + 2\,H^+_{(aq)} \rightarrow 2\,Na^+_{(aq)} + CO_{2(g)} + H_2O_{(l)}$$

In an experiment 0.200 mole of sodium carbonate powder and 0.100 mole of hydrochloric acid were placed in an open beaker.

The reaction was allowed to proceed to completion.

(a) On the axes below sketch graphs to illustrate the rate of disappearance of the sodium carbonate powder and the hydrochloric acid over time. Also sketch the rate of appearance of the carbon dioxide. Clearly label **all three** graphs. (3 marks)



(b) On the axes above add a new graph for $Na_2CO_{3(s)}$ (note: use a dotted line -----) showing the effect of using large crystals of sodium carbonate instead of the powder. Explain in terms of the Collision theory why this graph is the shape you have drawn. (3 marks)

Question 33 (9 marks)

Strontium hydroxide, $Sr(OH)_2$ is slightly soluble in water. The equilibrium established when solid $Sr(OH)_2$ is in contact with its saturated solution is represented by

$$Sr(OH)_{2(s)} \rightleftharpoons Sr^{2+}_{(aq)} + 2OH^-_{(aq)}$$

An equilibrium mixture of solid strontium hydroxide and its saturated solution is prepared. Four test tubes are set up, each containing some of the equilibrium mixture – a little white solid under a colourless solution. Each of the test tubes is treated as described below.

Complete the table below by indicating the direction of the expected shift in equilibrium following the changes stated in the table. Use Le Chatelier's theory to briefly explain how the shift was predicted.

Test tube	*Change*	*Direction of shift in equilibrium (write 'left', 'right' or no change)*	*Reasons for shift (or no change)*
A	A little solid strontium iodide is added and the mixture shaken		
B	A little concentrated hydrochloric acid is added and the mixture shaken		
C	A little solid sodium carbonate is added and the mixture shaken		

Question 34 **(7 marks)**

Consider the following process at equilibrium.

$$\mathbf{CH_{4(g)} + H_2O_{(l)} \rightleftharpoons 3\,H_{2(g)} + CO_{(g)}} \qquad \Delta H = +205 \text{ kJ/mol}$$

(a) Predict whether the following changes will increase, decrease or have no effect to the rate at which equilibrium is attained. (3 marks)

Change	*Effect*
Liquid water is added	
The pressure of CH_4 is increased	
The temperature is decreased to a new constant value	

(b) Predict whether the following changes will increase, decrease or have no effect on the equilibrium yield of the reaction and the value of equilibrium constant (K) (4 marks)

Change	*Effect on yield*	*Effect on equilibrium constant (K)*
The system is cooled to a new temperature		
Decreasing the volume of the system		

Question 35 **(6 marks)**

Complete the following table below by giving a brief description of a **chemical** test that could be used to distinguish between the pair of substances listed. State relevant observations relating to the test used for each of substance 1 and substance 2.

Substances to be distinguished		***Brief*** *description of chemical test (chemical equations are* ***not required)***	*Observation (write nvr if no visible change is observed)*	
Substance 1	***Substance 2***		***Substance 1***	***Substance 2***
0.5 mol/L sodium sulfate solution	0.5 mol/L sodium hydroxide solution			
solid iron (II) carbonate	solid iron (II) hydroxide			

Question 36 **(6 marks)**

Write **balanced ionic equations** and observations for each of the following. In each case describe in full what you would observe, including

- any colours
- odours
- precipitates (give colour)
- gases evolved (give colour or describe as colourless).

If no change is observed, you should state this.

(a) 20.0 mL of 0.5 $molL^{-1}$ calcium chloride solution is mixed with 100.0 mL of 1.0 $molL^{-1}$ lead (II) nitrate solution. (3 marks)

Equation	
Observation	

(b) Solid calcium hydroxide is added to dilute ammonium iodide solution. (3 marks)

Equation	
Observation	

END OF BOOKLET 2

Semester One Examination, 2013

Question/Answer Booklet 3

CHEMISTRY 3AB

3

Student name: ______________________________

Teacher: ______________________________

For each question, shade the box to indicate your answer.
Use **only** a blue or black **pen** to shade the boxes.

For example, if b is your answer: a ☐ b ■ c ☐ d ☐

If you make a mistake, place a cross through that square then shade your answer.
Do not erase or use correction fluid.

For example, if b is a mistake and d is your correct answer: a ☐ b ■ c ☐ d ■

If you then want to use your first answer b, cross out d and then circle b.

a ☐ b ■ c ☐ d ■

Marks will **not** be deducted for incorrect answers.

No marks will be given if more than one answer is completed for any question.

1	a ☐ b ☐ c ☐ d ☐
2	a ☐ b ☐ c ☐ d ☐
3	a ☐ b ☐ c ☐ d ☐
4	a ☐ b ☐ c ☐ d ☐
5	a ☐ b ☐ c ☐ d ☐

6	a ☐ b ☐ c ☐ d ☐
7	a ☐ b ☐ c ☐ d ☐
8	a ☐ b ☐ c ☐ d ☐
9	a ☐ b ☐ c ☐ d ☐
10	a ☐ b ☐ c ☐ d ☐

11	a ☐ b ☐ c ☐ d ☐
12	a ☐ b ☐ c ☐ d ☐
13	a ☐ b ☐ c ☐ d ☐
14	a ☐ b ☐ c ☐ d ☐
15	a ☐ b ☐ c ☐ d ☐

16	a ☐ b ☐ c ☐ d ☐
17	a ☐ b ☐ c ☐ d ☐
18	a ☐ b ☐ c ☐ d ☐
19	a ☐ b ☐ c ☐ d ☐
20	a ☐ b ☐ c ☐ d ☐

21	a ☐ b ☐ c ☐ d ☐
22	a ☐ b ☐ c ☐ d ☐
23	a ☐ b ☐ c ☐ d ☐
24	a ☐ b ☐ c ☐ d ☐
25	a ☐ b ☐ c ☐ d ☐

Section Three: Extended answer 40% (80 Marks)

This section contains six (6) questions. You must answer all questions. Write your answers in the spaces provided.

Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression. Lists or dot points are unlikely to gain full marks.

Final answers to calculations should be expressed to three (3) significant figures.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 70 minutes

See next page

Question 37 **(14 marks)**

A heater company produces a portable gas heater that can run on either butane gas (C_4H_{10}) cylinders or household supply of natural gas (CH_4).

i. The equation for the complete combustion of butane gas is as follows:

$$\mathbf{2\ C_4H_{10(g)} + 13\ O_{2(g)} \rightarrow 8\ CO_{2(g)} + 10\ H_2O_{(l)}}$$

In normal use, one gas cylinder containing 500.0g of butane will give exactly 150 hours of use.

ii. The equation for the complete combustion of natural gas is as follows:

$$\mathbf{CH_{4(g)} + 2\ O_{2(g)} \rightarrow CO_{2(g)} + 2\ H_2O_{(l)}}$$

The methane from the household supply system is delivered at 200.0 kPa pressure and at a rate of 2.40 L per hour.

Assume a temperature of 20.0^{o}C throughout.

(a) Which gas (butane or methane) requires the most mass (in grams) of oxygen **per hour** to sustain complete combustion of the butane during normal operation? (show all working) (7 marks)

(b) Calculate the mass of water produced during 10.0 hours of normal operation when running on the butane gas cylinders. (2 marks)

On the first of July 2012, the Australian government imposed a carbon tax level of \$23.0 per tonne of carbon dioxide emitted on a number of manufacturing industries (1 tonne = 1×10^6 g).

Like many other industries, this heater company immediately announced a price increase of \$5.00 on this portable heater model to help it 'pay for the new tax'.

(c) Assuming the heater is operating from a household methane gas system and is used for 100 days a year, calculate the total mass of carbon dioxide produced by this portable heater over a period of 10 years (life expectancy of a heater) and state whether this price increase is fair for the consumers. (5 marks)

Question 38 **(13 marks)**

The preparation of sulfuric acid from iron pyrites (FeS_2) can be represented by the following series of reactions:

$$\mathbf{4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2}$$

$$\mathbf{2SO_2 + O_2 \rightleftharpoons 2SO_3} \qquad \textbf{(Contact Process)}$$

$$\mathbf{SO_3 + H_2O \rightarrow H_2SO_4}$$

(a) How many moles of sulfuric acid would be produced from **ONE** mole of iron pyrites? (1 mark)

(b) What mass of iron pyrites would be needed to produce 123 tonne (1 tonne = 1 x 10^6 g) of sulfuric acid if the Contact process only converts 92.5% of the sulfur dioxide and the other reactions achieve 100% conversion? (6 marks)

Sulfurous acid (H_2SO_3) can be produced by the dissolving of sulfur dioxide gas in liquid water under the right conditions.

$$SO_{2(g)} + H_2O_{(l)} \rightleftharpoons H_2SO_{3(aq)} \qquad \textit{(Reaction 1)}$$

Sulfurous acid partially ionises in solution

$$H_2SO_{3(aq)} \rightleftharpoons H^+_{(aq)} + HSO_3^-{}_{(aq)}$$

$$HSO_3^-{}_{(aq)} \rightleftharpoons H^+_{(aq)} + SO_3^{2-}{}_{(aq)}$$

(c) If ***Reaction 1*** above is slightly exothermic then predict the effect of the following changes on the pH of the solution: (3 marks)

Change to equilibrium	***Effect on pH (write increase, decrease or no change)***
The temperature of the system is increased to a new constant value	
Finely powdered $CaCO_{3(s)}$ is added	
A little concentrated nitric acid is added	

(d) The equilibrium constant (K) value for the ionisation of a 1.0 $molL^{-1}$ sulfurous acid solution is 1.40×10^{-2} at 25^oC and atmospheric pressure and the equilibrium constant value for the ionisation of a 1.0 $molL^{-1}$ hydrogensulfite (HSO_3^-) solution is 6.30×10^{-8} at the same conditions.

Which of these two solutions (sulfurous acid or hydrogen sulfite) would be a stronger electrolyte at 25^oC? Explain your answer. (3 marks)

Question 39 **(14 marks)**

A local water sample was analysed and found to contain a high concentration of magnesium and calcium ions.

Common identification tests for the presence of these ions include the precipitation reaction of these ions with sodium hydroxide or sodium carbonate solutions.

(a) Write relevant ionic equations to show the precipitation of magnesium and calcium ions from the water by dilute sodium hydroxide solution. (2 marks)

Equation 1: ____

Equation 2: ____

(b) Draw a Lewis electron dot diagram for $Ca(OH)_2$ (1 mark)

(c) A 5.00 mL water sample has a magnesium ion concentration of 12.1 mgL^{-1} **and** a calcium ion concentration of 5.01 mgL^{-1}. Calculate the mass of calcium ions and magnesium ions in the water sample. (2 marks)

A scientist attempts to remove all magnesium and calcium ions from a polluted lake using precipitation reaction. In a small trial, he adds 50.0 g of sodium carbonate decahydrate ($Na_2CO_3.10H_2O$) to 100.0 L of polluted water.

The relevant chemical equations for the precipitation reactions are as follow:

$$\mathbf{Mg^{2+}_{(aq)} + CO_3^{2-}{}_{(aq)} \rightarrow MgCO_{3(s)}}$$

$$\mathbf{Ca^{2+}_{(aq)} + CO_3^{2-}{}_{(aq)} \rightarrow CaCO_{3(s)}}$$

The polluted water from the lake was found to contain 2.50 g of calcium ions and 1.50 g of magnesium ions.

(d) Calculate the amount of carbonate ions required to remove all the calcium and magnesium ions from the water and hence determine whether the use of 50.0 g sodium carbonate decahydrate is enough for the total removal of these ions. (6 marks)

(e) From your answer to part (d),

- if the amount of sodium carbonate decahydrate is not enough, determine the mass of additional sodium carbonate decahydrate that needs to be added.

OR

- if the amount of sodium carbonate decahydrate is in excess, determine the mass of unreacted sodium carbonate decahydrate. (2 marks)

Question 40 **(15 marks)**

Both perchloric acid, $HClO_4$ and barium hydroxide, $Ba(OH)_2$ are substances that fully ionise/dissociate when in water solution.

(a) Calculate the pH of a 50.0 mL solution of 0.0250 $molL^{-1}$ perchloric acid. (1 mark)

(b) Calculate the pH of a 0.0250 $molL^{-1}$ solution of barium hydroxide. (3 marks)

(c) In an experiment, a student dissolves 0.215 g of solid $Ba(OH)_2$ in 50.0 mL of 0.025 $molL^{-1}$ perchloric acid. Calculate the final pH of the mixture. (6 marks)

(d) Assuming the mass of 1.00 L of the final solution mixture is 1.02 kg, determine the concentration of barium ions (in **parts per million)** at the end of the reaction. (2 marks)

The Arrhenius equation for the ionisation of perchloric acid in water is as follows:

$$HClO_{4(aq)} \rightarrow H^+_{(aq)} + ClO_4^-{}_{(aq)}$$

(e) What would be the expected colour of universal indicator in a solution mixture containing 10.0 mL of 0.105 mol/L perchloric acid ($HClO_4$) and 10.0 mL of 0.105 mol/L ammonia? Explain using a suitable Lowry-Bronsted equation. (2 marks)

Colour: ____

Equation: ____

(f) Would you expect a 1.0 molL^{-1} solution of $HClO_4$ to be a stronger or weaker electrolyte than a 1.0 molL^{-1} solution of CH_3COOH? Explain. (2 marks)

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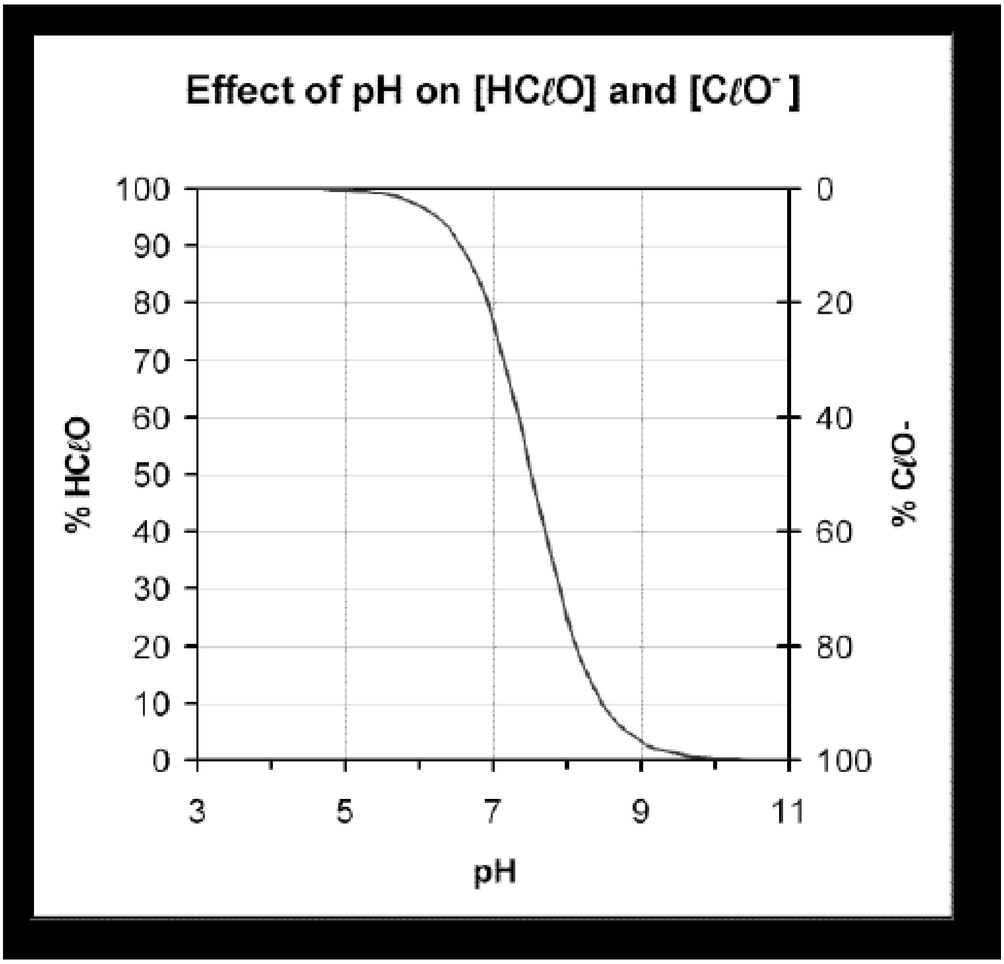
Question 41 **(17 marks)**

Common chemicals used to control algae and bacterial growth in swimming pools include hypochlorous acid HClO and the hypochlorite ion ClO^-.

The equilibrium chemical equation between HClO and ClO^- ion in water is

$$H_2O_{(l)} + HClO_{(aq)} \rightleftharpoons H_3O^+_{(aq)} + ClO^-_{(aq)}$$

The equilibrium is pH dependent and is very sensitive in the range of pH 7 to 8.



(a) Examine the graph above and state the effect of pH on the concentration of $HClO_{(aq)}$ and $ClO^-_{(aq)}$ ions. (1 mark)

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(b) State the pH at which the concentration of $HClO_{(aq)}$ equals the concentration of $ClO^-_{(aq)}$. (1 mark)

____

(c) Use Le Chatelier's principle to help explain the shape of the graph. Include in your explanation the effect of low **and** high pH on

i. the position of equilibrium and

ii. the concentration of $HClO$ and ClO^-. (4 marks)

(d) Write the equilibrium expression (K) for the reaction: $\mathbf{H_2O_{(l)} + HClO_{(aq)} \rightleftharpoons H_3O^+_{(aq)} + ClO^-_{(aq)}}$ (1 mark)

See next page

(e) The equilibrium constant (K) value for the reaction: $H_2O_{(l)} + HClO_{(aq)} \rightleftharpoons H_3O^+_{(aq)} + ClO^-_{(aq)}$ is 3.5×10^{-8} at 25°C. Would you expect the concentration of HClO to be greater, smaller or the same as the concentration of ClO^- ions at 25°C? Explain your answer. (2 marks)

Ammonia compounds and organic matter enter a swimming pool from many sources. Swimmers and bathers are major contributors with their bodies giving off saliva, sweat, urine and faecal matter.

The removal of ammonia via chemical reaction with HClO to form chloramine (NH_2Cl) is thus very important:

$$\mathbf{NH_{3(aq)} + HClO_{(aq)} \rightarrow NH_2Cl_{(aq)} + H_2O_{(l)}}$$

Chloramine is mildly toxic to bacteria, but tends to be irritating to eyes and skin, so it is an unwelcome addition to the pool. Fortunately, chloramine reacts with each other, as

$$\mathbf{NH_2Cl_{(aq)} + NH_2Cl_{(aq)} \rightarrow N_{2(g)} + 3\,HCl_{(aq)}}$$

(f) An Olympic swimming pool was analysed for ammonia in the water and found to have an ammonia level of 0.175 mg/L. If the swimming pool contains 2,500,000 litres of water then determine the number of moles of hydrochloric acid produced from the reactions above. (2 marks)

(g) Hydrochloric acid can lower the pH of the water in swimming pools. Water with low pH will dissolve plaster and grout in swimming pools and corrode metal products.

Sodium hydrogencarbonate ($NaHCO_3$) or sodium carbonate (Na_2CO_3) powder is often used to remove the hydrochloric acid that lowers the pH of swimming pools and spas.

The **unbalanced** equation for the reaction between sodium carbonate and hydrochloric acid is:

$$\mathbf{Na_2CO_{3(s)} + H^+_{(aq)} \rightarrow H_2O_{(l)} + CO_{2\,(g)} + Na^+_{(aq)}}$$

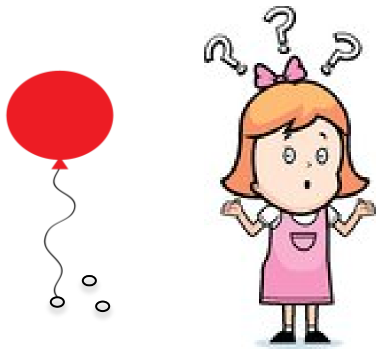
In a swimming pool water testing procedure, a technician recorded a pH reading of 6.50 for a swimming pool containing 2,500,000 litres of water. In order to return the water back to neutral (i.e. a pH reading of exactly 7), he added 44.1 g of 95.0% pure sodium carbonate (Na_2CO_3) to the swimming pool.

Carry out necessary calculations to determine whether the added amount of Na_2CO_3 was able to achieve the desired result (i.e. a final water pH of 7). (6 marks)

Question 42 **(7 marks)**

Christine buys some balloons to fill with helium (He) for a party. To stop them floating away, she decides to tie small metal rings to each balloon.

The following information is relevant to the questions that follow.



Mass of a balloon	***= 3.60 g***
Mass of string tied to each balloon	***= 1.50 g***
Mass of each metal ring	***= 1.20 g***
Volume of balloon when inflated	***= 15.0 L***
Pressure of gas in inflated balloon:	***= 103 kPa***
Temperature	***= 25.0° C***

(a) Determine the maximum number of moles of gas that can be held by the balloon at the conditions stated above. (1 mark)

(b) Calculate the **minimum** number of metal rings she will need to tie to each balloon to stop it floating away.

Useful information:

i. The balloon will float upwards if the total mass of the balloon, string, metal rings and the helium gas inside the balloon is less than the mass of air (and nothing else!) removed from a fully inflated balloon.

ii. One (1) mole of air has a mass of 28.8 grams. (6 marks)

END OF PAPER